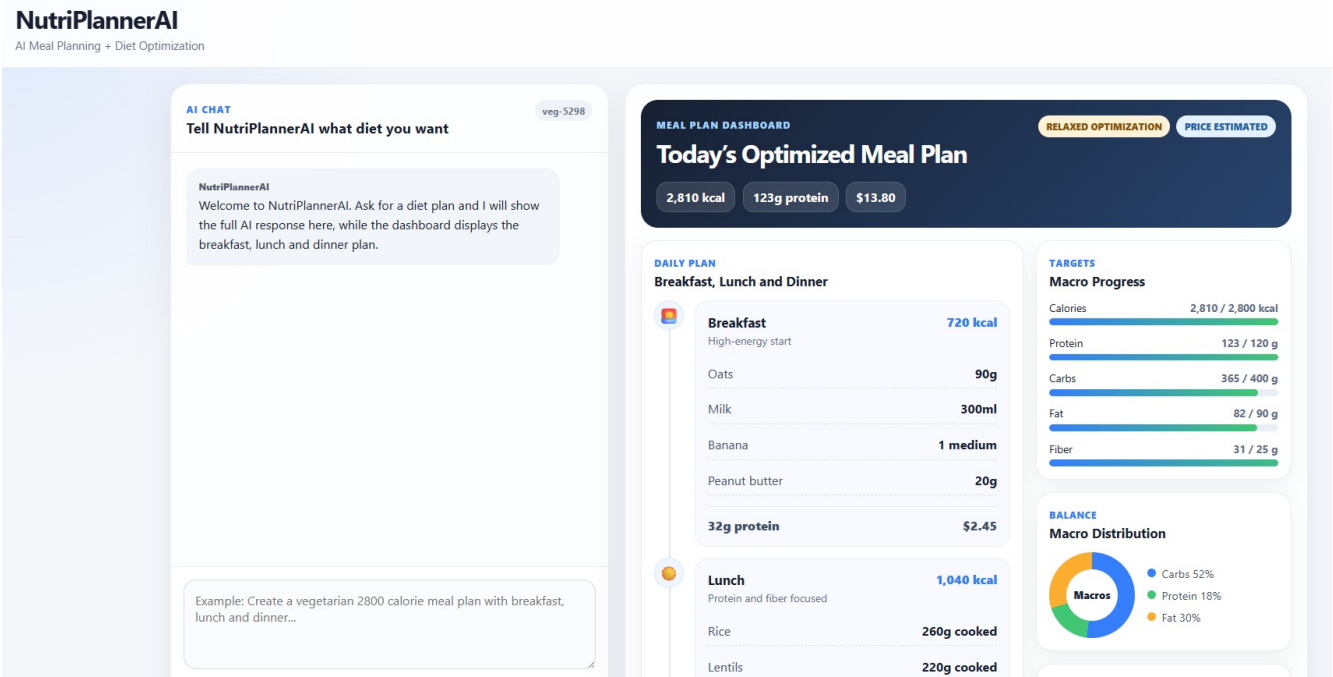


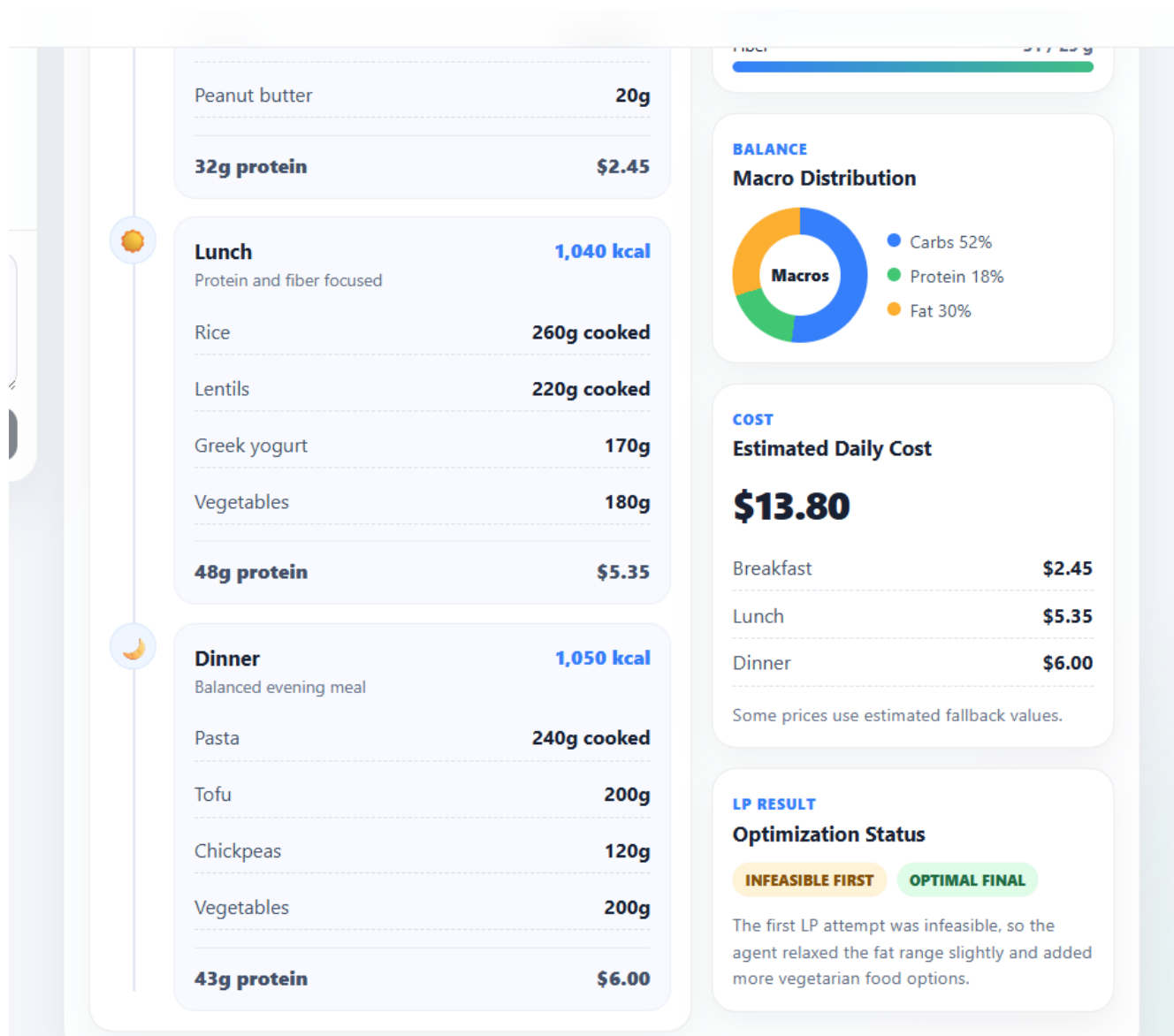
NutriPlannerAI setup

Go to nutriplannerai-frontend folder

npm run dev

Initial UI





I tried this prompt:

Create a vegetarian 2800 calorie meal plan with at least 120g protein. Use affordable foods, get nutrition from OpenFoodFacts, estimate prices if needed, run optimization, and give me breakfast, lunch, dinner, daily totals, and tools used.

Response

Conversation ID: veg-5298

I created a vegetarian meal plan that meets your calorie and protein requirements, with a focus on affordability. The total calories are slightly below the target, but the protein goals are achieved. Here's the breakdown:

****Breakfast:****

- Quaker Large Flake Oats: 150g
- Whole Milk 3.25%: 250g
- Banana: 200g

- Peanut Butter: 100g

****Lunch:****

- Tofu: 109g
- Lentils: 291g

****Dinner:****

- Paneer: 100g
- Cottage Cheese: 100g

****Daily Totals:****

- Calories: 2650 kcal
- Protein: 134.7 g
- Carbs: 250 g
- Fat: 100 g
- Fiber: 36.98 g
- Estimated cost: \$6.30

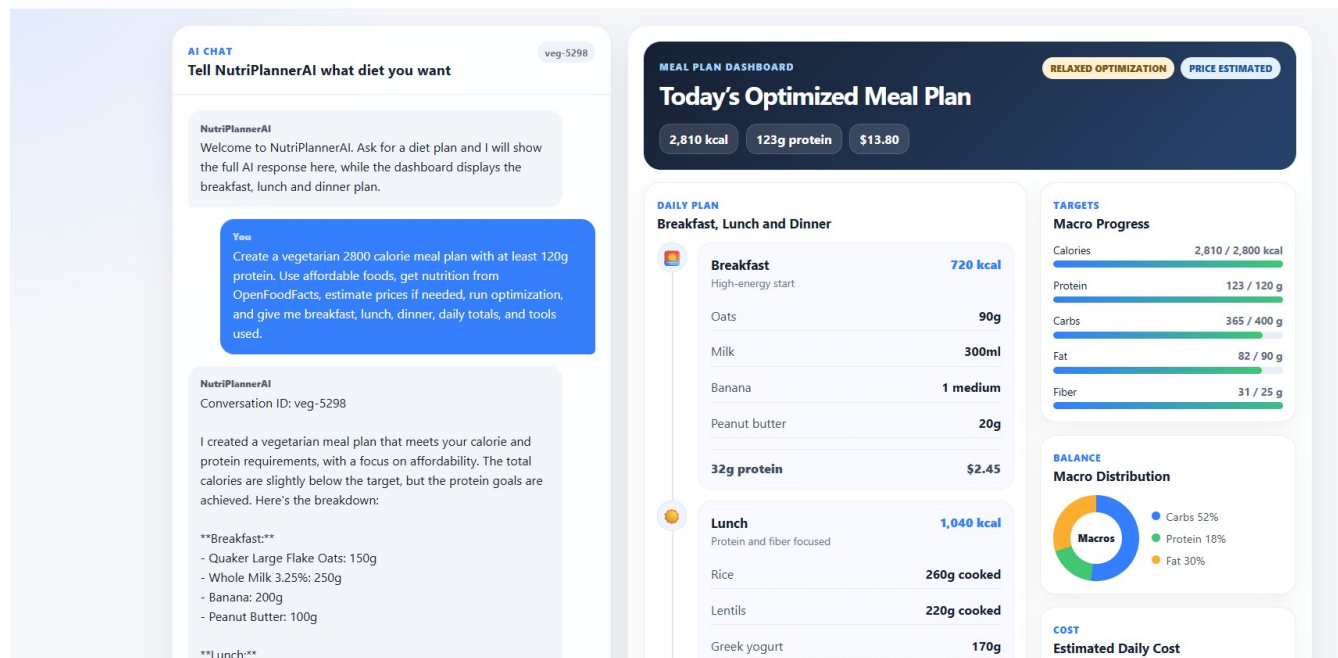
****Tools Used:****

- NutriOpt optimization tool for calculating the meal plan.
- Saved food items for nutritional data and pricing.

This meal plan provides a good balance of macronutrients, while keeping the overall cost reasonable. If you'd like to make adjustments or need help with anything else, feel free to ask!

NutriPlannerAI

AI Meal Planning + Diet Optimization



Today's Optimized Meal Plan

2,810 kcal

123g protein

\$13.80

DAILY PLAN

Breakfast, Lunch and Dinner



Breakfast

720 kcal

High-energy start

Oats 90g

Milk 300ml

Banana 1 medium

Peanut butter 20g

32g protein

\$2.45



Lunch

1,040 kcal

Protein and fiber focused

Rice 260g cooked

Lentils 220g cooked

Greek yogurt 170g

Vegetables 180g

48g protein

\$5.35

TARGETS

Macro Progress

Calories 2,810 / 2,800 kcal

Protein 123 / 120 g

Carbs 365 / 400 g

Fat 82 / 90 g

Fiber 31 / 25 g

BALANCE

Macro Distribution



COST

Estimated Daily Cost

\$13.80

Breakfast \$2.45

Peanut butter 20g

32g protein \$2.45

Lunch 1,040 kcal
Protein and fiber focused

Rice 260g cooked

Lentils 220g cooked

Greek yogurt 170g

Vegetables 180g

48g protein \$5.35

Dinner 1,050 kcal
Balanced evening meal

Pasta 240g cooked

Tofu 200g

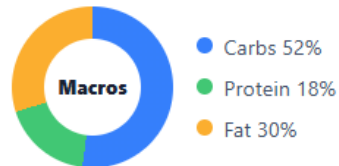
Chickpeas 120g

Vegetables 200g

43g protein \$6.00

BALANCE

Macro Distribution



COST

Estimated Daily Cost

\$13.80

Breakfast \$2.45

Lunch \$5.35

Dinner \$6.00

Some prices use estimated fallback values.

LP RESULT

Optimization Status

INFEASIBLE FIRST OPTIMAL FINAL

The first LP attempt was infeasible, so the agent relaxed the fat range slightly and added more vegetarian food options.

MariaDB

```
MariaDB [(none)]> use nutriopt_db;
Database changed
MariaDB [nutriopt_db]> show tables;
+-----+
| Tables_in_nutriopt_db |
+-----+
| agent_conversations |
| agent_messages |
| food_items |
+-----+
3 rows in set (0.002 sec)

MariaDB [nutriopt_db]> select * from food_items;
+-----+
| id | barcode | brand | calories_per100g | carbs_per100g | cost_per100g | currency | fat_per100g | fiber_per100g | generic_name | latest_price | max_grams | min_grams | name |
+-----+
| 1 | 055577105026 | Quaker | 389 | 66 | 0.499 | USD | 7 | 8 | oats | 0.499 | 500 | 100 | Quaker Large Flake |
| 2 | 064420001234 | Neilson | 61 | 5 | 0.162 | USD | 3.25 | 0 | milk | 0.162 | 1000 | 250 | Whole Milk 3.25% |
| 3 | 060383000111 | No Name | 143 | 1 | 0.665 | USD | 10.6 | 0 | eggs | 0.665 | 700 | 100 | Large Eggs |
| 4 | 000000000101 | Generic | 165 | 0 | 1.2 | CAD | 3.6 | 0 | chicken breast | 12 | 350 | 100 | Chicken Breast |
| 5 | 000000000102 | Generic | 130 | 28 | 0.25 | USD | 0.3 | 0.5 | rice | 0.25 | 500 | 100 | White Rice Cooked |
| 6 | 000000000103 | Generic | 89 | 23 | 0.169 | USD | 0.3 | 2.6 | banana | 0.169 | 200 | 100 | Banana |
| 7 | 068100091234 | Kraft | 588 | 20 | 0.599 | USD | 50 | 6 | peanut butter | 0.599 | 500 | 100 | Peanut Butter |
```

```
MariaDB [nutriopt_db]> select * from food_items;
+-----+
| id | barcode | brand | calories_per100g | carbs_per100g | cost_per100g | currency | fat_per100g | fiber_per100g | generic_name | latest_price | max_grams | min_grams | name |
+-----+
| 1 | 055577105026 | Quaker | 389 | 66 | 0.499 | USD | 7 | 8 | oats | 0.499 | 500 | 100 | Quaker Large Flake | |
| 2 | 064420001234 | Neilson | 61 | 5 | 0.162 | USD | 3.25 | 0 | milk | 0.162 | 1000 | 250 | Whole Milk 3.25% |
| 3 | 060383000111 | No Name | 143 | 1 | 0.665 | USD | 10.6 | 0 | eggs | 0.665 | 700 | 100 | Large Eggs |
| 4 | 000000000101 | Generic | 165 | 0 | 1.2 | CAD | 3.6 | 0 | chicken breast | 12 | 350 | 100 | Chicken Breast |
| 5 | 000000000102 | Generic | 130 | 28 | 0.25 | USD | 0.3 | 0.5 | rice | 0.25 | 500 | 100 | White Rice Cooked |
| 6 | 000000000103 | Generic | 89 | 23 | 0.169 | USD | 0.3 | 2.6 | banana | 0.169 | 200 | 100 | Banana |
| 7 | 068100091234 | Kraft | 588 | 20 | 0.599 | USD | 50 | 6 | peanut butter | 0.599 | 500 | 100 | Peanut Butter |
| 8 | 060383001234 | No Name | 65 | 14 | 0.399 | USD | 0.4 | 3.3 | mixed vegetables | 0.399 | 500 | 100 | Frozen Mixed Vegetables |
| 9 | 3017624010701 | Ferrero | 539 | 57.5 | 0 | USD | 30.9 | 0 | Chocolate Hazelnut Spread | 0 | 1000 | 100 | Nutella |
| 10 | 400 | 2023-10-01 | Open Food Facts | 6.3 | 9 | USD | 6.9 | 10.1 | Oats | 0 | 1000 | 100 | Oats |
| 11 | 1000 | 2023-10-01 | Generic | 16.9 | 1kg | 0 | USD | 3.6 | 0 | Chicken Breast | 0 | 1000 | 100 | Chicken Breast |
| 12 | 1000 | 2023-10-01 | Generic | 123 | 25.6 | 0 | USD | 0.9 | 1.8 | Brown Rice (cooked) | 0 | 1000 | 100 | Brown Rice |
| 13 | 1000 | 2023-10-01 | Generic | 143 | 1.1 | 0 | USD | 9.5 | 0 | Eggs | 0 | 1000 | 100 | Eggs |
| 14 | 1000 | 2023-10-01 | Generic | 588 | 20 | 0 | USD | 50 | 6 | Peanut Butter | 0 | 1000 | 100 | Peanut Butter |
| 15 | 000000000104 | Generic | 370 | 6 | 0.5 | USD | 1.5 | 2 | Tofu | 0.5 | 1000 | 100 | Tofu |
| 16 | 000000000105 | Generic | 116 | 20 | 0.3 | USD | 0.4 | 4 | Lentils | 0.3 | 1000 | 100 | Lentils |
| 17 | 000000000106 | Generic | 312 | 4 | 2 | USD | 24 | 0 | Paneer | 2 | 1000 | 100 | Paneer |
| 18 | 000000000107 | Generic | 98 | 4 | 0.8 | USD | 4 | 0 | Cottage Cheese | 0.8 | 1000 | 100 | Cottage Cheese |
```